

Beamer⁺

Interactive Presentations

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aliasing.py

Program

```
1 def add_item(items, item):
2     items.append(item)
3
4 shopping = ["milk"]
5 also = shopping
6 add_item(also, "eggs")
7 print(shopping)
```

Shell

Python 3.11.3 (Pyodide)

```
>>> shopping
['milk', 'eggs']
>>> for i in range(2):
...     print(i)
0
1
```

```
>>> 1/0
```

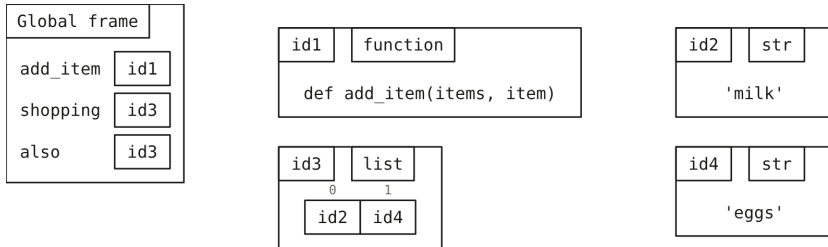
Traceback (most recent call last):

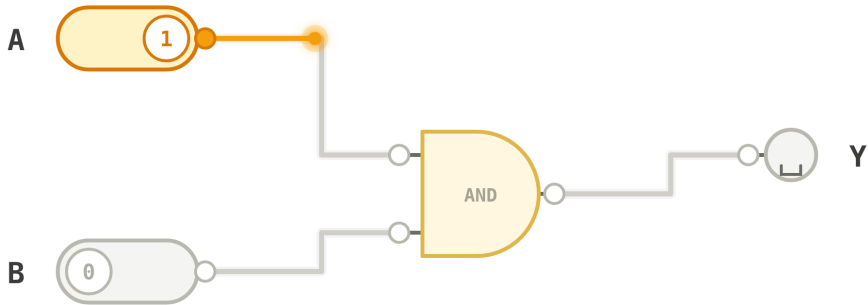
File "<stdin>", line 1, in <module>

ZeroDivisionError: division by zero

Python failed to load. Check the network connection, then press Restart.

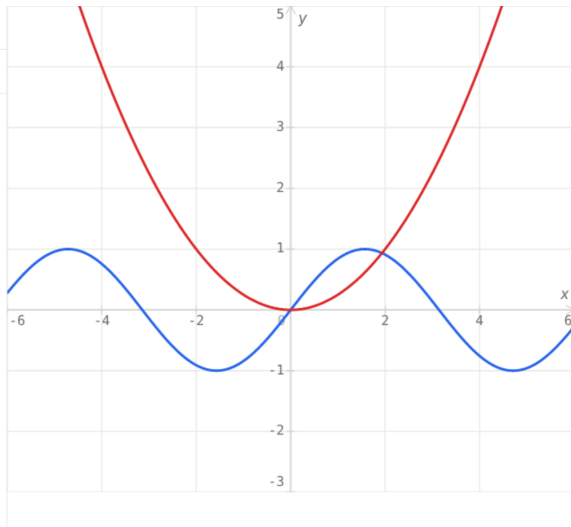
Memory model (step 6)





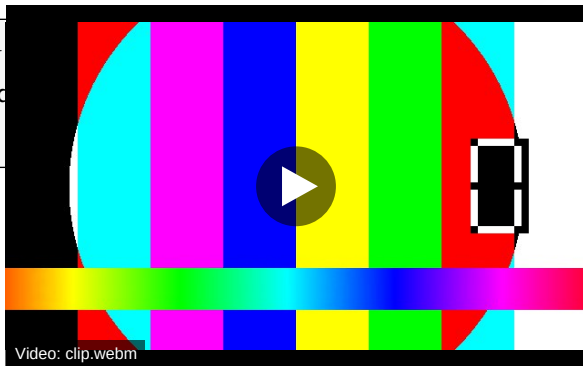
● $\sin(x)$

● $x^2/4$



Beamer+
embedded

ides with
ation, and



videos/audio

supported formats include .mp4, .mov, .mpeg, .mp3, .wav

example usage: demonstrating Newton's 2nd law in a Physics course



acceleration is
nt of

Audio: tone.mp3

try clicking!

24 responses

Which data structure gives $O(1)$ average lookup?

A

Linked list

13%

3 votes



Hash table

54%

13 votes

C

Binary search tree

25%

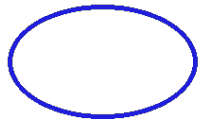
6 votes

D

Sorted array

8%

2 votes



What is one thing you're still unsure about?

18 responses

Hash collisions (×3)

When to use a tree vs a hash table

Big-O of amortised operations (×2)

Recursion (×2)

Why resizing doubles the array — café naïve “quotes” -> arrows $\leq \geq \pi$

Proof on slide 14

Pointers

Dynamic programming (×4)

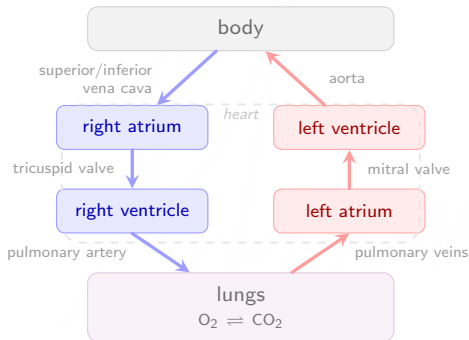
Graph traversal order

Heaps

models

supported formats include .glp**b**

example usage: explaining blood flow through the heart in an Anatomy course



blood flow through
(deoxygenated blood / oxygenated blood)

Calculator
Interactive widget

try rotating!

demo.ipynb

Demo notebook

Some *markdown* text with math x^2 .

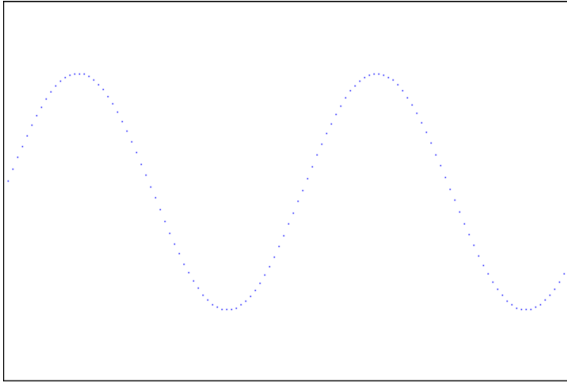
In [1]

```
import numpy as np  
print(np.arange(5))
```

```
[0 1 2 3 4]
```

In [2]

```
import matplotlib.pyplot as plt  
plt.plot(np.sin(np.linspace(0,10)))
```



In [3] `1/0`**ZeroDivisionError: division by zero**In [4] `x0 = 0*0`
`x0`

0

In [5] `x1 = 1*1`
`x1`

1

In [6] `x2 = 2*2`
`x2`

4

```
In [7] x3 = 3*3  
x3
```

9

```
In [8] x4 = 4*4  
x4
```

16

```
In [9] x5 = 5*5  
x5
```

25

```
In [10] x6 = 6*6  
x6
```

36

```
In [11] x7 = 7*7  
x7
```

49

```
In [12] x8 = 8*8  
x8
```

64

```
In [13] x9 = 9*9  
x9
```

81

```
In [14] x10 = 10*10  
x10
```

100

In [15]

```
x11 = 11*11  
x11
```

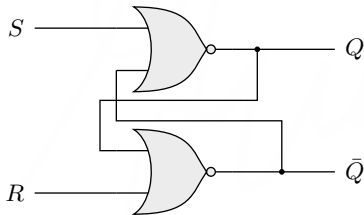
```
121
```

widgets

supported formats include `.html`

example usage: explaining latches in a Computer Organization class

A **latch** stores one bit of state using cross-coupled gate feedback.



cross-coupled NOR gates
inputs are *active-high*

S	R	Q	$\bar{Q}$	state
0	0	Q	$\bar{Q}$	hold
1	0	1	0	set
0	1	0	1	reset
1	1	0	0	forbidden

truth table for an SR latch
implemented with NOR gates

widgets

supported formats include `.html`

example usage: explaining latches in a Computer Organization class

A **late**

Never run poll: is $P = NP$?

A

True

B

False

learn more

visit the link below to learn more

`chandra-gummaluru.github.io/post/beamer-plus`